

1815**Multicylinder music box**

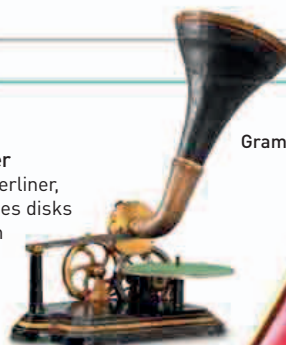
First produced in Switzerland in 1815, these contain a rotating cylinder covered in spikes that pluck the teeth of a steel comb. In 1862, a system that allows cylinders to be changed to play different melodies is invented.



Multicylinder music box

1888**Gramophone player**

Invented by Emile Berliner, the gramophone uses disks of shellac. They can be copied numerous times from a brass master without any loss of quality.



Gramophone

1857**Phonautograph**

Édouard-Léon Scott invents the first device able to record sound but it is unable to play the sound back.

1876**Player piano**

The automatic-playing piano becomes popular when it is shown at an exhibition. It operates by an electromagnet and contains a paper music roll.



Player piano

1877**Edison's phonograph**

Thomas Edison's phonograph is the first device able to play sounds back as well as record them. The vibrations of sound are funneled through a horn and recorded on a cylinder covered in tin foil.



horn concentrates sound for recording and amplifies it for playback

THE STORY OF SOUND RECORDING

RECORDING SOUND WAS AN ANCIENT DREAM OF MANKIND BUT IT ONLY BECAME A REALITY DURING THE 19TH CENTURY

Little more than a century ago, the only music most people heard was performed live. The advent of technology to record sound and play it back has not only transformed the way we listen to music but has also had other applications, including broadcasting, film-making, and sound archiving.

Frenchman Édouard-Léon Scott's phonautograph of 1857 was the first device capable of recording sound, using a moving needle to trace a line on a carbon-coated surface. In 1877, American Thomas Edison invented the phonograph—the first device that could record sound and play it



back. Early sound recording machines worked mechanically—they captured sound by playing into a horn to make the sound vibrations move a needle to engrave scratches on a disk or cylinder.

In the 1920s, sound recording entered the electrical era with the invention of microphones. Sound was soon being reproduced electrically

Edison and his phonograph

The phonograph was invented by Thomas Edison in 1877. It recorded sounds in grooves embossed in tin foil wrapped around a cylinder. Later cylinders were made from wax-coated cardboard.

using electromagnet-driven loudspeakers with improved sound quality and volume.

After 1945, sound recording of music took off with vinyl

records that played back at 33 or 45 revolutions per minute (rpm) (early records played back at 78 rpm). Magnetic devices that recorded sound as varying magnetic patterns on tape rather than physical grooves on a disk were also developed.

DIGITAL

The next big breakthrough was digital recording (see panel, right), which made for much more robust, practical systems. These include the first compact disks, as well as digital audio formats such as MP3, which have made it possible to store vast amounts of music on small devices, and download limitless music from the internet.

**“ ...I CAN TRANSPORT YOU
TO THE REALMS OF MUSIC. ”**

The first promotional message recorded on the Edison phonograph, 1877